

Probabilistic Retrieval Model

Introduction to Information Retrieval

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VSM and TF-IDF Weighting

- Combines TF and IDF to find the weight of terms

$$w(t, D) = \left(1 + \log\left(\frac{tf(t, D)}{|D|}\right)\right) \cdot \log \frac{|N|}{df(t)}$$

- For a query Q and document D , retrieval score $f(Q, D)$

$$score(Q, D) = \sum_{q \in Q} w(q, D)$$

- TF-IDF observations
 - ▶ Presence / absence most important (**term overlap**)
 - ▶ Term appearing more in a doc gets higher weight (**TF**)
 - ▶ First occurrence is more important (**log**)
 - ▶ Rare terms are more important (**IDF**)
 - ▶ Bias towards longer documents
 - ▶ Repeatability important in long documents

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- For a query

- Vector Space Model very heuristic in nature.
 - ▶ No notion of relevance.
 - ▶ Any weighting scheme, similarity measure can be used.
 - ★ Components not interpretable.
 - ★ More engineering rather than theory $\Rightarrow$ tweak, run, observe, repeat
 - ...

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Probabilistic Model

- Mathematical formalisation for relevant / non-relevant sets.

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Probabilistic Model

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Probability Ranking Principle

Probability Ranking Principle (PRP) - Robertson (1977)

PRP - a framework to define *'what is a good ranking?'*

- *If the response to a query is a ranking of the documents in the collection in order of **decreasing probability of relevance** ($P(R = 1|D, Q)$)*

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 - ▶ $P(R|D_{r1}) > P(R|D_{r2}) > P(R|D_{r3}) > P(R|D_{r4}) > \dots$

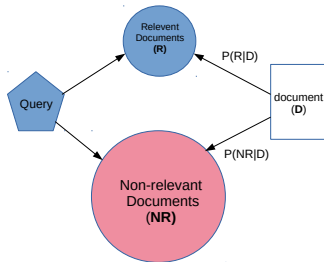
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 - Best possible accuracy can be achieved.

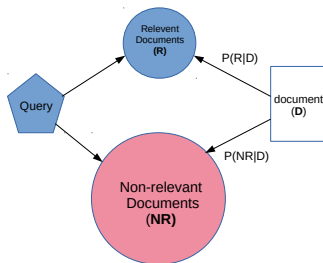
PRP concept

- Imagine IR as a classification problem:



PRP concept

- Imagine IR as a classification problem:



$$P(R|D) + P(NR|D) = 1$$

- D is relevant if $P(R|D) > P(NR|D)$ i.e. $\frac{P(R|D)}{P(NR|D)} > 0.5$.

Estimating the Probabilities

- Using Bayes Rule:

$$P(R|D) = \frac{P(D|R)P(R)}{P(D)}$$

$$P(NR|D) = \frac{P(D|NR)P(NR)}{P(D)}$$

- ▶ $P(D|R)$ - probability of observing D given that it is **relevant**.
 - ▶ $P(R)$ and $P(NR)$ - prior probability of relevance / non-relevance.
 - ▶ $P(D)$ - prior probability of the selection of the document.
 - ▶ $P(D|NR)$ - probability of observing D given that it is **non-relevant**.
- Classify a document as relevant if $P(R|D) > P(NR|D)$:

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- ▶ $P(D|R) > P(D|NR)$
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Estimating $P(D|R)$

assuming term independence

- Assume independence:

$$P(D|R) = \prod_{t \in D} P(t|R)$$

Probabilistic Retrieval Strategy

- Estimating the probabilities: **Binary Independence Model (BIM)** - the simplest model.
- Assumptions -
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- Assumptions -
 - ▶ Relevance of each document independent of relevance of other documents. (Not true).
 - ▶ Boolean model of relevance (a document either R or NR).
- Order documents by decreasing probability.

Binary Independence Model (BIM)

- **Binary**: documents represented as binary incidence vectors of terms
 - ▶ Document $D = \mathbf{D} = t_1^d, t_2^d, \dots, t_N^d$
 - ▶ $t_i^d = 1$ if term t_i present in document D
- **Independence**: terms occur in documents independently

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- **Independence**: terms occur in documents independently
- Query: binary term incidence vector $\mathbf{Q}$.

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$$P(R|\mathbf{D}, \mathbf{Q}) = O(R|D, Q) = \frac{P(R = 1|\mathbf{D}, \mathbf{Q})}{P(R = 0|\mathbf{D}, \mathbf{Q})}$$

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- Using term independence assumption

$$\frac{P(\mathbf{D}|R=1, \mathbf{Q})}{P(\mathbf{D}|R=0, \mathbf{Q})} = \prod_{i=1}^N \frac{P(t_i|R=1, \mathbf{Q})}{P(t_i|R=0, \mathbf{Q})}$$

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• Let

- ▶ $p_i = P(t_i=1|R=1, \mathbf{Q})$: probability that t_i **present** in **relevant** document
- ▶ $s_i = P(t_i=1|R=0, \mathbf{Q})$: probability that t_i **present** in **non-relevant** document
- ▶ $1 - p_i = P(t_i=0|R=1, \mathbf{Q})$: probability that t_i **absent** in **relevant** document
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	D is relevant	D is non-relevant
t_i present in D	p_i	s_i
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- Assumption $\rightarrow$ terms not occurring in query are equally likely to occur in relevant and non-relevant documents:
 - ▶ if $t_i \notin \mathbf{Q}$ then $p_i = s_i$.
 - ▶ Then we need only consider terms in the products that appear in the query, and so,

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$$\begin{aligned} RSV_D = \text{Score}(Q, D) &= \log\left(\prod_{i:d_i=q_i=1} \frac{p_i(1-s_i)}{s_i(1-p_i)}\right) \\ &= \sum_{i:d_i=q_i=1} \log \frac{p_i(1-s_i)}{s_i(1-p_i)} \end{aligned}$$

- $p_i = P(t_i = 1 | R = 1, \mathbf{Q})$: probability that t_i present in relevant document
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BIM

Scoring function

$$RSV_D = \text{Score}(Q, D) = \log \left(\prod_i p_i (1 - s_i) \right)$$

- $\text{weight}_{BIM} = \log \frac{N - df(i)}{df(i)}$

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$$RSV_D = \text{Score}(Q, D) = \log \left(\prod_{i=1}^n \frac{p_i(1 - s_i)}{\frac{N - df(i)}{df(i)}} \right)$$

- $weight_{BIM} = \log \frac{N - df(i)}{df(i)}$
- $\log \frac{N - df(i)}{df(i)} \approx \text{IDF}$ **How?**

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- ▶ s_i approximated by entire collection:

$$\text{score}(Q, D) = \sum_{t_i=1} \log \frac{0.5 \cdot (1 - \frac{df(i)}{N})}{\frac{df(i)}{N} \cdot (1 - 0.5)} = \sum_{t_i=1} \log \frac{N - df(i)}{df(i)}$$

Beyond Binary - The Basic Limitation of BIM

$$score_{BIM}(Q, D) = \sum_{q_i=1} \log \frac{N - df(i)}{df(i)}$$

Binary Term Presence

- Doesn't distinguish between 1 occurrence and 100.
- Real documents have term frequencies.

- Score documents based only on **collection statistics**.

Document A vs Document B

- **Query:** machine learning
- Doc A: machine x1, learning x1
- Doc B: machine x15, learning x20
- BIM treats them equally!

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Document A vs Document B

- **Query:** machine learning
- Doc A: machine x1, learning x1
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- BIM treats them equally!

- Score documents based only on collection statistics.
- Need some document specific features.
- Need to make sure length normalization.

The Best Match (BM) Models

A good term weighting is based on three principles

- inverse document frequency
- term frequency
- document length normalization

BIM covers only the first of these principles.

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A good term weighting is based on three principles

- inverse document frequency
- term frequency
- document length normalization

BIM covers only the first of these principles.

- Led to a series of experiments with the Okapi system¹ (developed for library catalogue search in City University, in London).

<https://sigir.org/files/museum/pub-26/59.pdf>

Beyond Binary Independence

- 1 Move beyond binary term presence
- 2 Add term frequency saturation
- 3 Introduce document length normalization
- 4 Maintain theoretical foundation

BM1: The Foundation of the BM Family

What is BM1?

- The **first "Best Match"** ranking function implemented in the Okapi system.
- Essentially the **Robertson/Spärck Jones (RSJ)**¹ weight applied to ad-hoc retrieval.
- Served as the baseline and starting point for all subsequent BM variants.

BM1 Scoring Formula

$$\text{BM1}(Q, d) = \sum_{t \in Q \cap d} \log \frac{N - df(t) + 0.5}{df(t) + 0.5}$$

Characteristics

- Uses only **IDF** for weighting
- No term frequency component
- No document length normalization

¹Robertson and Spärck Jones - JASIS 1976

BM11: Introducing Simple Term Frequency

BM11 Innovation

- Added a term frequency component with length normalization.
- Maintained the probabilistic foundation while addressing a key BIM limitation.

BM11 Scoring Formula

$$\text{BM11}(Q, d) = \sum_{t \in Q \cap d} \left(\frac{tf(t, d)(1 + k_1)}{tf(t, d) + k_1 \cdot \frac{\text{len}(d)}{\text{avgdl}}} \right) \cdot \log \frac{N - df(t) + 0.5}{df(t) + 0.5}$$

Key Features

- **Non-binary** term weighting
- Simple TF saturation: $\frac{tf}{tf+1}$
- TF component ranges from 0 to 1

Improvements over BM1

- Can distinguish between documents with different term frequencies
- Rewards documents where query terms appear more often

BM15 Model

BM15 Innovation

- Another attempt to incorporate term frequency **without length normalization**.

BM15 Scoring Formula

$$\text{BM15}(Q, d) = \sum_{t \in Q \cap d} \left(\frac{tf(t, d)(1 + k_1)}{tf(t, d) + k_1} \right) \cdot \log \frac{N - df(t) + 0.5}{df(t) + 0.5}$$

BM25 Model

BM11 TF component

$$\frac{tf(t,d)(1+k_1)}{tf(t,d)+k_1 \cdot \frac{len(d)}{avgdl}}$$

BM15 TF component

$$\frac{tf(t,d)(1+k_1)}{tf(t,d)+k_1}$$

- BM25: combination of the BM11 and BM15.

BM25 Model

BM11 TF component

$$\frac{tf(t,d)(1+k_1)}{tf(t,d)+k_1 \cdot \frac{len(d)}{avgdl}}$$

BM15 TF component

$$\frac{tf(t,d)(1+k_1)}{tf(t,d)+k_1}$$

- BM25: combination of the BM11 and BM15.
- The motivation was to combine the term frequency factors as follows:

$$B = \frac{tf \cdot (k_1 + 1)}{tf + k_1 \left[(1 - b) + b \frac{len(d)}{avgdl} \right]}$$

- ▶ where $b \rightarrow$ a constant with values in the interval $[0, 1]$.
- ▶ If $b = 0 \rightarrow$ BM15 term frequency factor.
- ▶ If $b = 1 \rightarrow$ BM11 term frequency factor.
- ▶ For values of b between 0 and 1 $\rightarrow$ provides a combination of BM11 with BM15.

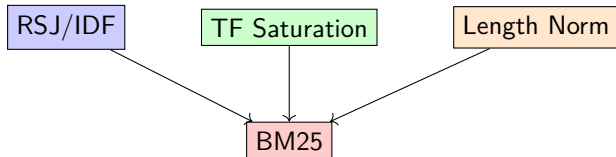
The Grand Synthesis: BM25

Combining All Components

- RSJ*-inspired IDF + Saturated TF + Length Normalization

$$\text{BM25}(t, d) = \frac{tf \cdot (1 + k_1)}{tf + k_1 \cdot B} \cdot \log \frac{N - n + 0.5}{n + 0.5}$$

$$\text{where } B = (1 - b) + b \cdot \frac{|d|}{\text{avgdl}}$$



* RSJ - Robertson/Spärck Jones

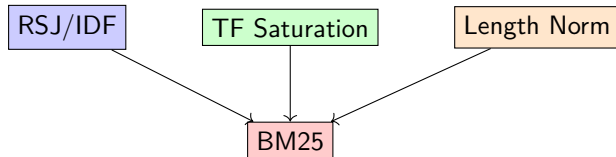
The Grand Synthesis: BM25

Combining All Components

- RSJ*-inspired IDF + Saturated TF + Length Normalization

$$weight_{BM25}(t, d) = \frac{tf(t, d)(1+k_1)}{tf(t, d) + k_1((1-b) + b \cdot \frac{|d|}{avgdl})} \cdot \log \frac{N - df(t) + 0.5}{df(t) + 0.5}$$

$$\text{where } B = (1 - b) + b \cdot \frac{|d|}{avgdl}$$



* RSJ - Robertson/Spärck Jones

What Does "Okapi BM25" Mean?

The Name

- **Okapi** → The system in which it was launched.
 - **BM** → "Best Match".
 - **25** → The 25th version tested in the Okapi project.
-
- Part of a family of models (BM1, BM11, BM15, BM25...).
 - Each variant tested in TREC conferences.
 - BM25 emerged as the most effective and robust.



Key Insight

BM25 wasn't a completely new invention. It was the **evolution and practical realization** of the RSJ probabilistic principle.

Okapi BM25 Model

Scoring

$$\begin{aligned} \text{score}(Q, d) &= \sum_{t \in Q \cap d} \text{weight}(t, d) \\ &= \sum_{t \in Q \cap d} \frac{tf(t, d)(1 + k_1)}{tf(t, d) + k_1(1 - b + b \cdot \frac{|d|}{\text{avgdl}})} \cdot \log \frac{N - df(t) + 0.5}{df(t) + 0.5} \end{aligned}$$

- $|d|$: length of document d
- avgdl : average document length
- k_1, b : constants; usually set empirically
- k_1 : controls term frequency scaling
 - ▶ $k_1 = 0$: binary model
 - ▶ k_1 large: raw term frequency
- b : controls document length normalization
 - ▶ $b = 0$: no length normalization
 - ▶ $b = 1$: fully normalization by average document length
- $k_1 = 1.2, b = 0.75$

Okapi BM25 Model

Scoring

$$\text{score}(Q, d) = \sum \text{weight}(t, d)$$

"BM25 and all that — a look back" - Stephen Robertson - SIGIR 2025

*BM25 seems to have been remarkably successful. Its development was a strange combination of analysis of what formal probabilistic models might suggest for the shapes of certain relationships, together with **completely adhoc, intuitive inventions of functions** that might imitate these shapes.*

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Disecting BM25

$$\text{score}(Q, d) = \sum_{t \in Q \cap d} \frac{tf(t, d)(1 + k_1)}{tf(t, d) + k_1(1 - b + b \cdot \frac{|d|}{\text{avgdl}})} \cdot \log \frac{N - df(t) + 0.5}{df(t) + 0.5}$$

- $\sum_{t \in Q \cap d}$: more words common with the query $\rightarrow$ better
- $tf(t, d)$: Repetition of query words $\rightarrow$ better
- $\log \frac{N - df(t) + 0.5}{df(t) + 0.5}$: Common words $\rightarrow$ less important

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 - ★ $b = 0$: no length normalization
 - ★ $b = 1$: fully normalization by average document length
- Repetitions less important than different query words, but more important if document length relatively longer.

BM25 vs. VSM

Why BM25 better than VSM TF-IDF

- Suppose $Q = \{\text{machine}, \text{learning}\}$
- Suppose we have 2 documents with term counts:
 - ▶ doc₁: learning 1024; machine 1
 - ▶ doc₂: learning 16; machine 8
 - ▶ Let: $\text{IDF}(\text{learning}) = \log_2(N/df) = 7$
 - ▶ Let: $\text{IDF}(\text{machine}) = \log_2(N/df) = 10$
- $\text{Rel}(\text{doc}_1)$?? $\text{Rel}(\text{doc}_2)$

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- VSM: $\log_2 tf * \log_2(N/df)$

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- $\text{Rel}(\text{doc}_1) < \text{Rel}(\text{doc}_2)$
- VSM: $\log_2 tf * \log_2(N/df)$
 - ▶ score(doc₁): $10 * 7 + 1 * 10 = 80$
 - ▶ score(doc₂): $4 * 7 + 3 * 10 = 58$

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- BM25: $\frac{(k_1+1)tf_i}{k_1 \left((1-b) + b \frac{|d|}{\text{avgdl}} \right) + tf_i} \log \frac{N}{df_i}, k_1 = 2, b = 0$

BM25 vs. VSM

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- BM25: $\frac{(k_1+1)tf_i}{k_1((1-b)+b\frac{|d|}{\text{avgdl}})+tf_i} \log \frac{N}{df_i}$, $k_1 = 2, b = 0$
 - ▶ score(doc₁): $(3 * 1024)/(2 + 1024) * 7 + (3 * 1/(2 + 1)) * 10 = 30.96$
 - ▶ score(doc₂): $(3 * 16)/(2 + 16) * 7 + (3 * 8/(2 + 8)) * 10 = 42.67$

Summary: Key Insights

Probabilistic Foundation

- PRP provides theoretical optimality guarantee
- BIM introduces probabilistic reasoning
- BM25 extends with practical considerations

BM25 Strengths

- Term frequency saturation
- Document length normalization
- Robust IDF formulation
- Few tunable parameters

Classic BM25 Recap

Standard BM25 Formula

$$\text{BM25}(Q, d) = \sum_{t \in Q \cap d} \frac{tf(t, d) \cdot (k_1 + 1)}{tf(t, d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{avgl}\right)} \cdot \log \frac{N - df(t) + 0.5}{df(t) + 0.5}$$

- k_1 : Term frequency saturation parameter (typically 1.2)
- b : Length normalization parameter (typically 0.75)
- $|d|$: Document length
- $avgl$: Average document length
- N : Total number of documents
- $df(t)$: Document frequency of term t

BM25 for Long Queries

BM25 for Long Queries

$$\text{BM25}(Q, d) = \sum_{t \in Q \cap d} \frac{tf(t, d) \cdot (k_1 + 1)}{tf(t, d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \cdot \log \frac{N - df(t) + 0.5}{df(t) + 0.5} \cdot \frac{(k_3 + 1) \cdot tf(t, Q)}{k_3 + tf(t, Q)}$$

Understanding the Components

- **First part:** Document TF with length normalization
- **Second part:** IDF component
- **Third part:** **Query TF weighting** with k_3

When to Use k_3

- **Appropriate for:** Paragraph-long information needs.

Key Parameters

- $tf(t, Q)$: Frequency of term t in query Q
- k_3 : Tuning parameter for query TF scaling
- **No query length normalization** (implicit $b = 0$)

Why No Query Length Normalization?

- Retrieval done with **single fixed query**
- Query length is constant during scoring
- Normalization would not affect ranking

BM25F - Field-based BM25 (Robertson, S., & Zaragoza, H. (2009))

Motivation

- Documents often have multiple fields (title, abstract, body, etc.)
- Different fields have different importance for relevance
- Classic BM25 treats all fields equally

BM25F Formula

$$\text{BM25F}(Q, d) = \sum_{t \in Q \cap d} \frac{tf_{\text{eff}}(t, d) \cdot (k_1 + 1)}{tf_{\text{eff}}(t, d) + k_1} \cdot \log \frac{N - df(t) + 0.5}{df(t) + 0.5}$$

where:

$$tf_{\text{eff}}(t, d) = \sum_{f \in \text{fields}} w_f \cdot \frac{tf(t, f)}{1 - b_f + b_f \cdot \frac{|f|}{\text{avg}|f|}}$$

BM25F - Implementation Example

Typical Field Weights

Field	Weight (w_f)	b_f
Title	2.0	0.75
Abstract	1.5	0.75
Body	1.0	0.75
Anchor text	1.2	0.50

- Higher weights for more important fields
- Different b parameters per field
- Effective for structured documents (web pages, academic papers)

BM25+ - Addressing TF Saturation (Lv, Y., & Zhai, C. (2011))

Problem with Classic BM25

- For very short documents, TF component can be too small
- Penalizes documents with low term frequencies
- May miss relevant short documents

BM25+ Solution

$$\text{BM25+}(Q, d) = \sum_{t \in Q \cap d} \frac{tf(t, d) \cdot (k_1 + 1)}{tf(t, d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} + \delta$$

where δ is a positive constant (typically 0.5-1.0)

- Adds a lower bound to term frequency component
- Ensures minimal contribution for each matching term
- Particularly helpful for short queries and documents

References and Further Reading

-  Robertson, S., & Zaragoza, H. (2009). The Probabilistic Relevance Framework: BM25 and Beyond
-  Lv, Y., & Zhai, C. (2011). Lower-bounding term frequency normalization
-  Lv, Y., & Zhai, C. (2014). Adaptive term frequency normalization for BM25
-  Trotman, A., & Keeler, J. (2016). Parameter-free BM25